

2020

DELHI

CBSE Solved Paper

Time allowed : 3 hours

Maximum Marks : 80

GENERAL INSTRUCTIONS:

Read the following instructions very carefully and strictly follow them:

- (i) The question paper comprises **three** sections **A, B** and **C**. There are 30 questions in the question paper. **All** questions are compulsory.
- (ii) **Section A** – all questions/or parts (question no. **1** to **14**) thereof in this section are **one** mark questions comprising **MCQ, VSA type** and **Assertion–Reason** type questions. They are to be answered in **one** word or in **one** sentence.
- (iii) **Section B** – question no. **15** to **24** are short answer type questions, carrying **three** marks each. Answer to these questions should not exceed **50** to **60** words.
- (iv) **Section C** – question no. **25** to **30** are long answer type questions, carrying **five** marks each. Answer to these questions should not exceed **80** to **90** words.
- (v) Answer should be brief and to the point. Also the above mentioned word limit be adhered to as far as possible.
- (vi) There is no overall choice in the question paper. However, an internal choice has been provided in some questions in each section. Only one of the choices in such questions have to be attempted.
- (vii) In addition to this, separate instructions are given with each section and question, wherever necessary.

SECTION-A

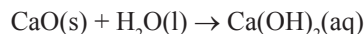
1. Name a cyclic unsaturated carbon compound. **(1 Mark)**
2. The change in magnetic field lines in a coil is the cause of induced electric current in it. Name the underlying phenomenon. **[OS] (1 Mark)**
Answer question numbers 3(a) to 3(d) and 4(a) to 4(d) on the basis of your understanding of the following paragraphs and the related studied concepts.
3. The growing size of the human population is a cause of concern for all people. The rate of birth and death in a given population will determine its size. Reproduction is the process by which organisms increase their population. The process of sexual maturation for reproduction is gradual and takes place while general body growth is still going on. Some degree of sexual maturation does not necessarily mean that the mind or body is ready for sexual acts or for having and bringing up children. Various contraceptive devices are being used by human beings to control the size of population.
 - (a) List two common signs of sexual maturation in boys and girls. **(1 Mark)**
 - (b) What is the result of reckless female foeticide? **(1 Mark)**

- (c) Which contraceptive method changes the hormonal balance of the body? **(1 Mark)**
 - (d) Write two factors that determine the size of a population. **(1 Mark)**
4. Human body is made up of five important components, of which water is the main component. Food as well as potable water are essential for every human being. The food is obtained from plants through agriculture. Pesticides are being used extensively for a high yield in the fields. These pesticides are absorbed by the plants from the soil along with water and minerals and from the water bodies. These pesticides are taken up by the aquatic animals and plants. As these chemicals are not biodegradable, they get accumulated progressively at each trophic level. The maximum concentration of these chemicals gets accumulated in our bodies and greatly affects the health of our mind and body.
 - (a) Why is the maximum concentration of pesticides found in human beings? **(1 Mark)**
 - (b) Give one method which could be applied to reduce our intake of pesticides through food to some extent. **(1 Mark)**
 - (c) Various steps in a food chain represent: **(1 Mark)**
 - (i) Food web
 - (ii) Trophic level
 - (iii) Ecosystem
 - (iv) Biomagnification

(d) With regard to various food chains operating in an ecosystem, man is a: **(1 Mark)**

- (i) Consumer
- (ii) Producer
- (iii) Producer and consumer
- (iv) Producer and decomposer

5. Calcium oxide reacts vigorously with water to produce slaked lime.



This reaction can be classified as:

- (A) Combination reaction (B) Exothermic reaction
- (C) Endothermic reaction (D) Oxidation reaction

Which of the following is a correct option? **(1 Mark)**

- (a) (A) and (C) (b) (C) and (D)
- (c) (A), (C) and (D) (d) (A) and (B)

OR

When hydrogen sulphide gas is passed through a blue solution of copper sulphate, a black precipitate of copper sulphide is obtained and the sulphuric acid so formed remains in the solution. The reaction is an example of a:

- (a) Combination reaction
- (b) Displacement reaction
- (c) Decomposition reaction
- (d) Double displacement reaction

6. In a double displacement reaction such as the reaction between sodium sulphate solution and barium chloride solution: **(1 Mark)**

- (A) exchange of atoms takes place
- (B) exchange of ions takes place
- (C) a precipitate is produced
- (D) an insoluble salt is produced

The correct option is:

- (a) (B) and (D) (b) (A) and (C)
- (c) only (B) (d) (B), (C) and (D)

7. Baking soda is a mixture of: **(1 Mark)**

- (a) Sodium carbonate and acetic acid
- (b) Sodium carbonate and tartaric acid
- (c) Sodium hydrogen carbonate and tartaric acid
- (d) Sodium hydrogen carbonate and acetic acid

8. The chemical formula for plaster of Paris is: **(1 Mark)**

- (a) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (b) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
- (c) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (d) $2\text{CaSO}_4 \cdot \text{H}_2\text{O}$

9. The laws of reflection hold true for: **(1 Mark)**

- (a) plane mirrors only (b) concave mirrors only
- (c) convex mirrors only (d) all reflecting surfaces

OR

When an object is kept within the focus of a concave mirror, an enlarged image is formed behind the mirror. This image is:

- (a) Real (b) Inverted
- (c) Virtual and inverted (d) Virtual and erect

10. At the time of short circuit, the electric current in the circuit: **(1 Mark)**

- (a) vary continuously (b) does not change
- (c) reduces substantially (d) increases heavily

OR

Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be:

- (a) 0.4 A (b) 0.6 A
- (c) 0.8 A (d) 1 A

11. Which one of the following is responsible for the sustenance of underground water? **[OS] (1 Mark)**

- (a) Loss of vegetation cover
- (b) Diversion for high water demanding crops
- (c) Pollution from urban wastes
- (d) Afforestation

12. Incomplete combustion of coal and petroleum: **[OS] (1 Mark)**

- (A) increases air pollution.
- (B) increases efficiency of machines.
- (C) reduces global warming.
- (D) produce poisonous gases.

The correct option is:

- (a) (A) and (B) (b) (A) and (D)
- (c) (B) and (C) (d) (C) and (D)

For question numbers 13 and 14, two statements are given one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

- (a) Both A and R are true and R is correct explanation of the Assertion.
- (b) Both A and R are true but R is not the correct explanation of the Assertion.
- (c) A is true but R is false.
- (d) A is false but R is true.

13. **Assertion (A):** Esterification is a process in which a sweet smelling substance is produced. **(1 Mark)**

Reason (R): When esters react with sodium hydroxide an alcohol and sodium salt of carboxylic acid are obtained.

- 14. Assertion (A):** In the process of nuclear fission, the amount of nuclear energy generated by the fission of an atom of uranium is so tremendous that it produces 10 million times the energy produced by the combustion of an atom of carbon from coal. **(1 Mark)**

Reason (R): The nucleus of a heavy atom such as uranium, when bombarded with low energy neutrons, splits apart into lighter nuclei. The mass difference between the original nucleus and the product nuclei gets converted to tremendous energy. **[OS]**

SECTION-B

- 15.** 1 g of copper powder was taken in a China dish and heated. What change takes place on heating? When hydrogen gas is passed over this heated substance, a visible change is seen in it. Give the chemical equations of reactions, the name and the color of the products formed in each case. **(3 Marks)**

- 16.** List the important products of the Chlor-alkali process. Write one important use of each. **(3 Marks)**

OR

How is washing soda prepared from sodium carbonate? Give its chemical equation. State the type of this salt. Name the type of hardness of water which can be removed by it?

- 17.** 3 mL of ethanol is taken in a test tube and warmed gently in a water bath. A 5% solution of alkaline potassium permanganate is added first drop by drop to this solution, then in excess. **(3 Marks)**

- How is 5% solution of KMnO_4 prepared?
- State the role of alkaline potassium permanganate in this reaction. What happens on adding it in excess?
- Write chemical equation of this reaction.

- 18.** A squirrel is in a scary situation. Its body has to prepare for either fighting or running away. State the immediate changes that take place in its body so that the squirrel is able to either fight or run? **(3 Marks)**

OR

Why is chemical communication better than electrical impulses as a means of communication between cells in a multi-cellular organism?

- 19.** Define the term pollination. Differentiate between self pollination and cross pollination. What is the significance of pollination? **(3 Marks)**
- 20.** What are homologous structures? Give an example. Is it necessary that homologous structures always have a common ancestor. Justify your answer. **[OS] (3 Marks)**
- 21.** Why is Tyndall effect shown by colloidal particles? State four instances of observing the Tyndall effect. **(3 Marks)**

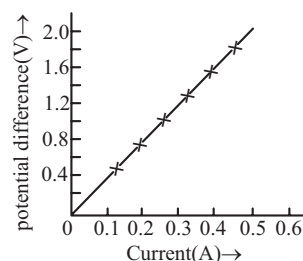
OR

Differentiate between a glass slab and a glass prism. What happens when a narrow beam of (i) a monochromatic light, and (ii) white light passes through

- (a) glass slab and (b) glass prism?

- 22.** Draw a labelled diagram to show
- reddish appearance of the sun at the sunrise or the sunset and **[OS]**
 - white appearance of the sun at noon when it is overhead. **(3 Marks)**

- 23.** A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph. **(3 Marks)**



- 24.** (a) Write the mathematical expression for Joule's law of heating.
(b) Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V. **(3 Marks)**

SECTION-C

- 25.** Carbon cannot reduce the oxides of sodium, magnesium and aluminium to their respective metals. Why? Where are these metals placed in the reactivity series? How are these metals obtained from their ores? Take an example to explain the process of extraction along with chemical equations. **(5 Marks)**
- 26.** The position of certain elements in the Modern Periodic Table are shown below.

Group →	1	2	3 to 12	13	14	15	16	17	18
↓ Period									
1	G								H
2	A			I			B		C
3		D			E				F

Using the above table answer the following questions giving reasons in each case: **[OS] (5 Marks)**

- Which element will form only covalent compounds?
- Which element is a non-metal with valency 2?
- Which element is a metal with valency 2?
- Out of H, C and F which has largest atomic size?
- To which family does H, C and F belong?

OR

Define atomic size. Give its unit of measurement. In the modern periodic table what trend is observed in the atomic radius in a group and a period and why is it so? **[OS]**

27. (a) Why is there a difference in the rate of breathing between aquatic organisms and terrestrial organisms? Explain.
- (b) Draw a diagram of human respiratory system and label-pharynx, trachea, lungs, diaphragm and alveolar sac on it. **(5 Marks)**

OR

- (a) Name the organs that form the excretory system in human beings.
- (b) Describe in brief how urine is produced in human body.
28. (a) What is the law of dominance of traits? Explain with an example. **(5 Marks)**
- (b) Why are the traits acquired during the life time of an individual not inherited? Explain. **[OS]**
29. Draw a ray diagram in each of the following cases to show the formation of image, when the object is placed: **(5 Marks)**

- (i) between optical centre and principal focus of a convex lens.
- (ii) anywhere in front of a concave lens.
- (iii) at 2F of a convex lens.

State the signs and values of magnifications in the above mentioned cases (i) and (ii).

OR

An object 4.0 cm in size, is placed 25.0 cm in front of a concave mirror of focal length 15.0 cm.

- (i) At what distance from the mirror should a screen be placed in order to obtain a sharp image?
- (ii) Find the size of the image.
- (iii) Draw a ray diagram to show the formation of image in this case.
30. (a) What is an electromagnet? List any two uses. **(5 Marks)**
- (b) Draw a labelled diagram to show how an electromagnet is made.
- (c) State the purpose of soft iron core used in making an electromagnet.
- (d) List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed.

EXPLANATIONS

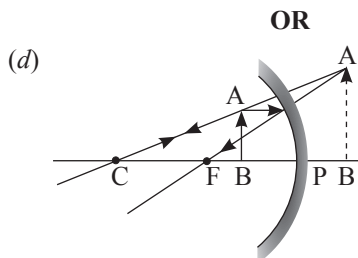
1. Cyclopropane is a cyclic unsaturated carbon compound. **(1 Mark)**
2. *Out of the Syllabus*
3. (a) Two common signs of sexual maturation in boys and girls are:
- Growth of hairs in particular body regions like the pubic area and underarms. **(½ Mark)**
 - Occurrence of acne. **(½ Mark)**
- (b) Reckless female foeticide refers to the illegal practice of aborting unborn female children, leading to an abnormal gender ratio between men and women. **(1 Mark)**
- (c) The oral contraceptive method works by changing the hormonal balance in the body, which in turn prevents ovulation in women who use these pills. **(1 Mark)**
- (d) Two key factors that determines the population size are: Birth rate and Death rate. **(1 Mark)**
4. (a) The maximum concentration of pesticides (accumulated at each trophic level by the process called biomagnification) are found in humans as they occupy the topmost trophic level in food chain. **(1 Mark)**

- (b) Washing food before consumption is a necessary step to ensure the removal of potentially harmful pesticides. **(1 Mark)**
- (c) (ii) Various steps in the food chain represent the trophic level, i.e., a specific position occupied by organism in the food chain based on their source of food. **(1 Mark)**
- (d) (i) With regard to various food chains operating in an ecosystem, man is a consumer, i.e., they depend on plants and other organisms for food. **(1 Mark)**
5. (d) Calcium hydroxide is formed when calcium oxide reacts with water. As a result, it is a combination reaction. It is an exothermic reaction because a lot of heat is released during the reaction, accompanied by a hissing sound. **(1 Mark)**

OR

- (d) It is double displacement reaction since cations and anions switches its positions and it occurs as:
- $$\text{H}_2\text{S} + \text{CuSO}_4 \rightarrow \text{CuS} + \text{H}_2\text{SO}_4 \quad \textbf{(1 Mark)}$$
6. (d) The reaction is: $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$
- Hence, exchange of ions takes place, a precipitate and an insoluble salt (BaSO_4) is produced. **(1 Mark)**

7. (c) Baking soda is mixture of sodium hydrogen carbonate and tartaric acid. (1 Mark)
8. (c) The chemical formula of Plaster of Paris is $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$. (1 Mark)
9. (d) The laws of reflection hold true for all reflecting surfaces. (1 Mark)



When the object is between the Pole and the Focus

Image is virtual and erect

(1 Mark)

10. (d) At the time of short circuit, the electric current in the circuit increases heavily. (1 Mark)

OR

- (d) As the two bulbs are connected in series, the current through both bulbs will be the same. (1 Mark)

11. (d) Out of the Syllabus

12. (b) Out of the Syllabus

13. (b) Esterification is the reaction of carboxylic acid with alcohol to produce the sweet-smelling chemical ester. When an ester interacts with sodium hydroxide, it produces an alcohol as well as the sodium salt of carboxylic acid. This is referred to as ester hydrolysis. (1 Mark)

14. (a) Out of the Syllabus

15. When 1g of copper powder is heated in a china dish, it produces a black material. Copper oxide (CuO) is the chemical formula for this material.

The reaction is:



Copper metal is created by passing hydrogen gas over heated Copper Oxide (CuO).

The black-coloured CuO is reduced, yielding brown-coloured Cu metal.



16. The important products of the Chlor-Alkali processes are:

1. Sodium Hydroxide (½ Mark)
2. Chlorine gas (½ Mark)
3. Hydrogen gas (½ Mark)

Uses:

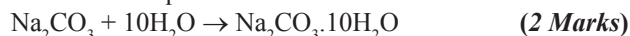
1. NaOH : It is utilised in the production of detergents and soaps, as well as paper and artificial fibres. (½ Mark)
2. Cl_2 : Used for making Bleaching powder, Pesticides, and CFCs. (½ Mark)

3. H_2 : Used for making ammonia for fertilisers, margarine and fuels. (½ Mark)

OR

Recrystallisation of sodium carbonate gives washing soda.

Its chemical equation is:



It is a basic salt. (½ Mark)

It is used for removing permanent hardness of water.

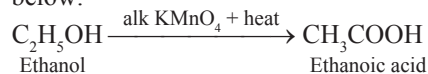
(½ Mark)

17. (i) Preparation of 5% Potassium permanganate: 5% solution of KMnO_4 is prepared by dissolving 5 grams of KMnO_4 in 100 mL of water. (1 Mark)

- (ii) Role of alkaline potassium permanganate:

1. Alkaline potassium permanganate acts as a strong oxidising agent in this reaction since it adds oxygen to alcohol and converts it into an acid. (½ Mark)
2. When it is added, the colour immediately fades because coloured permanganate ions of potassium permanganate are consumed in order to oxidise ethanol. (½ Mark)
3. When too much is added, the colour does not change because there is no more alcohol available for the reaction. (½ Mark)

- (iii) The chemical equation of this reaction is shown below:



(½ Mark)

18. When faced with a need to fight or escape, squirrels undergo immediate physiological changes in their bodies. These changes can be summarized as follows:

Adrenaline hormone release: Increasing heart and breathing rates.

Blood pressure and pupil changes: Alterations in blood pressure along with dilation of the pupil, facilitating their ability to run effectively.

Activation of sympathetic nervous system: Hypothalamus activates the sympathetic nervous system, leading to the release of adrenaline from the adrenal gland.

Pause in digestive and reproductive activities: During these situations, the squirrel temporarily halts digestive and reproductive processes.

All these changes made squirrels to be active while running or fighting like situations. (3 Marks)

OR

Chemical communication is a more effective means of cell-to-cell communication in multicellular organisms compared to electrical impulses due to the following reasons:

Chemical communication occurs through hormones, eliminating the need for specialized tissue like nervous tissue used in electrical impulses. (1½ Marks)

While electric communication is limited to specific regions connected by nerves, chemical communication is not region-specific and occurs throughout the body.

(1½ Marks)

19. The process by which transfer of pollen grains occurs from anther to stigma of the flower of the same species is known as pollination. (1 Mark)

Self-Pollination	Cross-Pollination
When the pollen grains transfer from anther to the stigma of the same flower, then this type of pollination is known as self pollination.	When the pollen grains transfer from anther of a flower to the stigma of the different flower then this type of pollination is called cross pollination.

(1 Mark)

Significance of pollination:

Pollen grains play a crucial role in fertilization by facilitating the transfer of male gametes to female eggs, leading to seed production and the subsequent generation of new offspring. Hence, pollination is important to develop new offsprings.

(1 Mark)

20. Out of the Syllabus

21. The Tyndall effect is shown by colloidal particles because of their small size and ability to scatter light. When light passes through a colloidal solution, the particles in the solution scatter the light, making it visible. This happens because the size of the particles is similar to the wavelength of visible light, causing the light to scatter in all directions. This scattering effect is known as the Tyndall effect.

(1 Mark)

Four instances of observing the Tyndall effect are:

1. When sunlight passes through a cloudy sky, the scattered light makes the clouds visible. (½ Mark)
2. When a beam of light is passed through a smoky room, the smoke particles scatter the light, making the beam visible. (½ Mark)
3. When a flashlight is shone through a glass of milk, the scattered light makes the milk appear white or opaque. (½ Mark)
4. When a laser beam is passed through a colloidal solution, the scattered light can be seen as a diffuse glow. (½ Mark)

OR

Glass slab and glass prism are both transparent objects made of glass, but they differ in their shapes and the way they refract light.

The main difference between a glass slab and a glass prism is that a prism can disperse light into its constituent colors, while a glass slab cannot.

(1 Mark)

- (i) When a narrow beam of monochromatic light passes through:

- (a) Glass prism: It has deviated from the actual path but the direction of the incident ray and emergent ray of light are not parallel to each other

(½ Mark)

- (b) Glass slab: It deviates from the actual path but the direction of the incident ray and emergent rays remain parallel to each other. (½ Mark)

- (ii) When a narrow beam of white light passes through:

- (a) Glass prism: White light is divided into its seven vibrant colors, violet, indigo, blue, green, yellow, orange, and red (VIBGYOR), and the directions of the incident and emerging rays are not parallel to others. (½ Mark)

- (b) Glass slab: There is no breaking of white light itself into the constituent color. Furthermore, the incident and emerging rays of light are parallel to each other. (½ Mark)

22. (i) Out of the Syllabus

- (ii) At noon, the sun is overhead in the sky and the light coming from the sun travels a relatively shorter distance through the atmosphere to reach the earth. In this case the blue light is not scattered much. As the light coming from the overhead sun contains almost all its component colours in the right proportion, the sun appears white to us at noon. (1½ Marks)

23. The nichrome cable's V-I graph is a straight line graph. Hence, it follows the ohm's law.

$$V = IR,$$

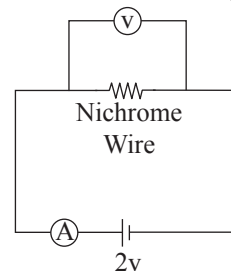
where, R is the resistance. V is the voltage. I is the current.

(1 Mark)

The resistance of a wire will be-

$$\Rightarrow R = \frac{V}{I} = \frac{0.4}{0.1} = 4 \Omega$$

Thus, the circuit diagram for the given case is,



(2 Marks)

24. (a) Joule's law of heating states that the amount of heat produced when a current flows through a conductor is directly proportional to the square of the current (I), the resistance (R) of the conductor, and the time (t) for which the current flows. The mathematical expression for Joule's law of heating is:

$$H = I^2 R t$$

(1 Mark)

- (b) Time (t) = 2hr = $2 \times 60 \times 60 = 7200\text{sec}$
 Charge (Q) = 96000C
 Potential difference (V) = 40V

$$\text{Current (I)} = \frac{Q}{T}$$

$$I = \frac{96000}{7200} = 13.33\text{A} \quad (1 \text{ Mark})$$

Now, heat generated

$$H = V \times I \times t$$

$$H = 40 \times 13.33 \times 7200$$

$$H = 3.8 \times 10^6 \text{ J} \quad (1 \text{ Mark})$$

25. Carbon cannot reduce the oxides of sodium, magnesium, and aluminium because their reactivity exceeds that of carbon and also they have more affinity for oxygen.

(1 Mark)

In the reactivity series, these metals are placed at the top above carbon. (1 Mark)

Electrolytic reduction (Electrolysis) is a process used to recover metals with high reactivity from their ores, such as sodium, magnesium, and aluminium. (1 Mark)

For example, sodium is obtained by the electrolysis of its molten chloride. The metal is deposited at the cathode (the negatively charged electrode), whereas, chlorine is liberated at the anode (the positively charged electrode).

(1 Mark)

The reactions are –



26. Out of the Syllabus

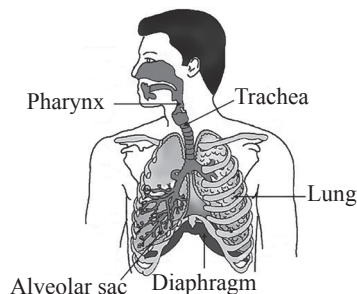
27. (a) The rate of respiration varies between aquatic and terrestrial organisms due to the following reasons:

Aquatic organisms, such as fishes, extract oxygen from water through their gills since it is present in a dissolved state. (1 Mark)

Since the amount of dissolved oxygen in water is lower compared to the oxygen in the gaseous form in air, aquatic organisms need higher breathing rates to fulfill their oxygen requirements. (1 Mark)

They breathe at a faster rate than humans to meet their body's demand for oxygen. (½ Mark)

- (b) (2½ Marks)



OR

- (a) The human excretory system comprises the following organs:

Two kidneys: Positioned on either side of the spine at the level of the liver, kidneys play a vital role in blood purification. They exist in pairs. (½ Mark)

A urinary bladder: Resembling a sac, the urinary bladder stores urine until it is expelled. It receives urine from the ureters, which are individually connected to each kidney. (1 Mark)

Urethra: A tube-like structure that originates from the urinary bladder. In females, it is shorter, while in males, it is longer. In males, the urethra serves as a common pathway for both sperm and urine. (1 Mark)

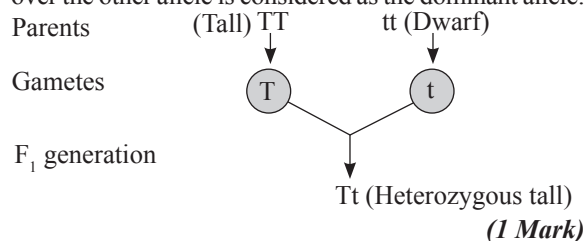
- (b) The primary role of the kidneys is to filter blood and excrete waste in the form of urine. The kidneys consist of a fundamental structure called nephrons, which play a crucial role in the filtration process. Nephrons have two main components: (1 Mark)

Bowman's capsule: Shaped like a cup, Bowman's capsule consists of a coiled tube of blood capillaries that aid in the filtration process. (½ Mark)

Renal tubule: The filtrate obtained from Bowman's capsule enters the renal tubule. During its passage, certain substances, such as amino acids, glucose, water and salts, are reabsorbed and the remaining filtrate is called urine is collected in the collecting duct. (½ Mark)

Subsequently, the filtered fluid moves through a duct and travels towards the urinary bladder, where it is stored until excreted from the body as urine. (½ Mark)

28. (a) The law of dominance of traits states that in a hybrid condition, the allele that expresses its characteristics over the other allele is considered as the dominant allele. Parents (Tall) TT tt (Dwarf)



(1 Mark)

In this case one parent has a homozygous dominant allele and the other parent has a homozygous recessive allele, their gametes will carry the dominant and recessive traits respectively. (1 Mark)

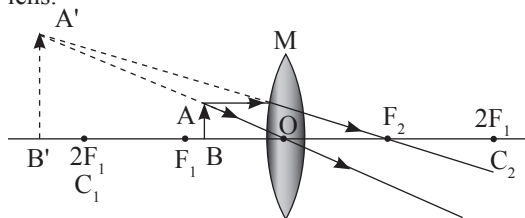
After fertilization, the first generation F₁ would have heterozygous conditions among the both alleles the 'T' character expresses over the 't' allele.

Hence we can say that here 'T' is dominant over 't'.

(½ Mark)

- (b) Out of the Syllabus

29. (i) Between optical centre and principal focus of a convex lens.

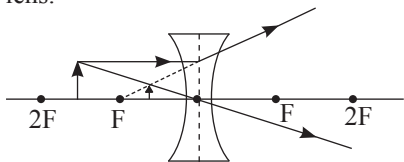


(1 Mark)

Here, the image formed is virtual, magnified and erect. So, magnification is positive and greater than one.

(1 Mark)

- (ii) When an object is placed anywhere in front of a concave lens.

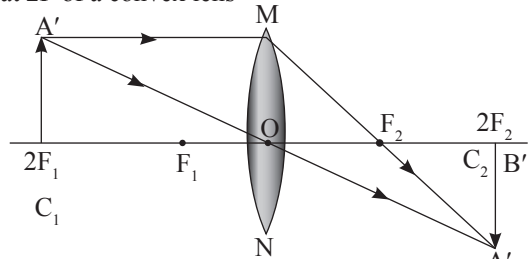


(1 Mark)

Here, the image formed is diminished, virtual and erect. So, magnification is positive and less than one.

(1 Mark)

- (iii) at $2F$ of a convex lens



(1 Mark)

OR

Given;

Focal length (f) = -15 cm

Object distance (u) = -25 cm

Object height (h_o) = 4 cm

- (i) From mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-15} - \frac{1}{-25} = \frac{-5+3}{75}$$

$$v = \frac{-75}{2} = -37.5 \text{ cm}$$

(1 Mark)

Hence, the screen must be placed at a distance of 37.5 in front of a mirror.

(½ Mark)

- (ii) We can find out the size of image using magnification formula

$$m = \frac{-v}{u} = \frac{h_i}{h_o} \Rightarrow \frac{-\left(\frac{-75}{2}\right)}{-25} = \frac{h_i}{4}$$

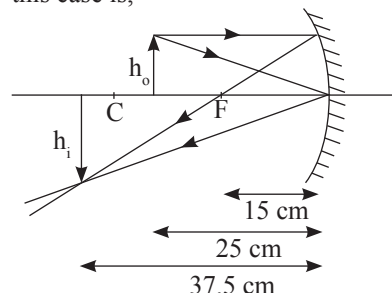
$$h_i = \frac{-75}{2 \times 25} \times 4 = -6 \text{ cm}$$

(1 Mark)

Here, the image is below principal axis.

(½ Mark)

- (iii) The ray diagram showing the formation of image in this case is,



(2 Marks)

30. (a) An electromagnet is a magnet that is created by passing an electric current through a coil of wire wrapped around a ferromagnetic core such as iron. The magnetic field generated by the current in the coil magnetizes the core.

(1 Mark)

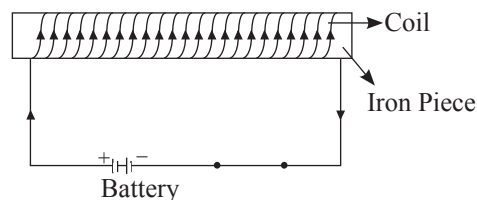
Two uses of electromagnets are:

- In electric motors: Electromagnets are used in electric motors to convert electrical energy into mechanical energy.
- In MRI machines: Electromagnets are used in MRI (magnetic resonance imaging) machines to create a strong magnetic field that aligns the protons in the body's tissues.

(½ Mark)

(½ Mark)

- (b) Diagram to show how an electromagnet made:



(1 Mark)

- (c) The purpose of a soft iron core used in making an electromagnet is to increase the magnetic field produced by the electromagnet. When a current flows through the wire coil wrapped around the soft iron core, it generates a magnetic field. The soft iron core has high magnetic permeability, which means it can amplify the magnetic field produced by the current in the coil. As a result, the electromagnet becomes stronger and more efficient.

(1 Mark)

- (d) If the material of an electromagnet is fixed, there are two ways to increase its strength;

- (i) Increase the number of turns in the coil.

(½ Mark)

- (ii) Increase the current flowing through the coil.

(½ Mark)